

C2 vs C3 — Per-Question Comparison by Ambiguity Type

v3 substrate, Claude Sonnet 4.6 backend. Author: Aaron Tan. Run: local Neo4j Desktop (policev3), 2026-07-27 — companion to the canonical Kaya run. Source: results/qual_records_claude-sonnet_v3_localdb.json via experiments/analyze_qualitative.py.

Legend: **green** = the generated query matches the benchmark's default interpretation (EX correct). **Red** = it does not (EX wrong); the small note under the query gives the failure_mode (valid_non_default = a different valid reading than the default; wrong_result = matches no gold interpretation).

Schema ambiguity

n=20 — 4 fixed by C3, 0 regressed, 3 both_correct, 13 both_wrong

Question	C2 — schema_grounded	C3 — disambiguation_enhanced
Q-041 Show people connected to the Northbridge robbery.	<pre>MATCH (loc:Location {suburb:'Northbridge'})<-[:OCCURRED_AT]- (i:Incident {crime_type:'robbery'})<-[:SUSPECTED_OF]- (p:Person)-[:ASSOCIATED_WITH]->(p2:Person) RETURN DISTINCT p2.name</pre> <i>X wrong_result</i>	<pre>MATCH (p:Person)-[r]->(i:Incident)-[:OCCURRED_AT]- >(l:Location {suburb:'Northbridge'}) WHERE i.crime_type = 'robbery' RETURN DISTINCT p.name</pre> <i>✓ matches default</i>
Q-042 Who is involved in the Cannington homicide?	<pre>MATCH (i:Incident {crime_type:'homicide'})-[:OCCURRED_AT]- >(l:Location {suburb:'Cannington'}) WITH i MATCH (p:Person)- [:SUSPECTED_OF WITNESSED VICTIM_OF INVESTIGATES]->(i) RETURN DISTINCT p.name</pre> <i>✓ matches default</i>	<pre>MATCH (p:Person)- [:SUSPECTED_OF WITNESSED VICTIM_OF INVESTIGATES]->(i:Incident {crime_type:'homicide'})-[:OCCURRED_AT]->(l:Location {suburb:'Cannington'}) RETURN DISTINCT p.name</pre> <i>✓ matches default</i>
Q-043 List everyone linked to the fraud on St Georges Terrace.	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'fraud'})-[:OCCURRED_AT]->(l:Location {address:'St Georges Terrace'}) RETURN DISTINCT p.name</pre> <i>X wrong_result</i>	<pre>MATCH (p:Person)-[r]->(i:Incident)-[:OCCURRED_AT]- >(l:Location {address:'St Georges Terrace'}) WHERE i.crime_type = 'fraud' RETURN DISTINCT p.name</pre> <i>X wrong_result</i>
Q-044 Who took part in the assault in Scarborough?	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'assault'})-[:OCCURRED_AT]->(l:Location {suburb:'Scarborough'}) RETURN DISTINCT p.name</pre> <i>X valid_non_default</i>	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'assault'})-[:OCCURRED_AT]->(l:Location {suburb:'Scarborough'}) RETURN DISTINCT p.name</pre> <i>X valid_non_default</i>
Q-045 Show the people associated with the drug offence on Lake Street.	<pre>MATCH (l:Location {address:'Lake Street'})<-[:OCCURRED_AT]- (i:Incident {crime_type:'drug_offence'})<-[:SUSPECTED_OF]- (p:Person) RETURN DISTINCT p.name</pre> <i>X wrong_result</i>	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'drug_offence'})-[:OCCURRED_AT]->(l:Location {address:'Lake Street'}) RETURN DISTINCT p.name</pre> <i>X wrong_result</i>
Q-046 Who is connected to the Midland robbery?	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'robbery'})-[:OCCURRED_AT]->(l:Location {suburb:'Midland'}) RETURN DISTINCT p.name</pre> <i>X valid_non_default</i>	<pre>MATCH (l:Location {suburb:'Midland'})<-[:OCCURRED_AT]- (i:Incident {crime_type:'robbery'})<-[r]-(p:Person) RETURN DISTINCT p.name</pre> <i>✓ matches default</i>
Q-047 Which people have a role in the burglary in Fremantle?	<i>X invalid_query</i>	<i>X invalid_query</i>
Q-048 Who is named in connection with the homicide?	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'homicide'}) RETURN DISTINCT p.name</pre> <i>X valid_non_default</i>	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'homicide'}) RETURN DISTINCT p.name</pre> <i>X valid_non_default</i>
Q-049 Show all parties to the fraud at 100 Hay Street.	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'fraud'})-[:OCCURRED_AT]->(l:Location {address:'100 Hay Street'}) RETURN DISTINCT p.name</pre> <i>X valid_non_default</i>	<pre>MATCH (l:Location {address:'100 Hay Street'})<- [:OCCURRED_AT]-(i:Incident {crime_type:'fraud'})<- [:SUSPECTED_OF WITNESSED VICTIM_OF INVESTIGATES]-(p:Person) RETURN DISTINCT p.name</pre> <i>✓ matches default</i>

Question	C2 — schema grounded	C3 — disambiguation enhanced
Q-050 Who is involved in the assault in Northbridge?	MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'assault'})-[:OCCURRED_AT]->(l:Location {suburb:'Northbridge'}) RETURN DISTINCT p.name <i>X valid_non_default</i>	MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'assault'})-[:OCCURRED_AT]->(l:Location {suburb:'Northbridge'}) RETURN DISTINCT p.name <i>X valid_non_default</i>
Q-051 List people tied to the drug offence in Midland.	MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'drug_offence'})-[:OCCURRED_AT]->(l:Location {suburb:'Midland'}) RETURN DISTINCT p.name <i>X valid_non_default</i>	MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'drug_offence'})-[:OCCURRED_AT]->(l:Location {suburb:'Midland'}) RETURN DISTINCT p.name <i>X valid_non_default</i>
Q-052 Who is connected to the traffic incident?	MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'traffic'}) RETURN DISTINCT p.name <i>X valid_non_default</i>	MATCH (p:Person)-[r]->(i:Incident {crime_type:'traffic'}) RETURN DISTINCT p.name <i>✓ matches default</i>
Q-053 Who has a connection to the burglary in Victoria Park?	MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'burglary'})-[:OCCURRED_AT]->(l:Location {suburb:'Victoria Park'}) RETURN DISTINCT p.name <i>X valid_non_default</i>	MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'burglary'})-[:OCCURRED_AT]->(l:Location {suburb:'Victoria Park'}) RETURN DISTINCT p.name <i>X valid_non_default</i>
Q-054 Which persons are linked to the William Street robbery?	MATCH (l:Location {address:'William Street'})<-[:OCCURRED_AT]-(i:Incident {crime_type:'robbery'})<-[:SUSPECTED_OF]-(p:Person) RETURN DISTINCT p.name <i>X wrong_result</i>	MATCH (loc:Location {address:'William Street'})<-[:OCCURRED_AT]-(i:Incident {crime_type:'robbery'})<-[r]-(p:Person) RETURN DISTINCT p.name <i>X wrong_result</i>
Q-055 Who is involved in the St Georges Terrace fraud — the suspects or the officers?	MATCH (l:Location {address:'St Georges Terrace'})<-[:OCCURRED_AT]-(i:Incident {crime_type:'fraud'}) WITH i OPTIONAL MATCH (suspect:Person)-[:SUSPECTED_OF]->(i) OPTIONAL MATCH (officer:Person)-[:INVESTIGATES]->(i) RETURN DISTINCT suspect.name AS suspect, officer.name AS officer <i>X wrong_result</i>	MATCH (l:Location {address:'St Georges Terrace'})<-[:OCCURRED_AT]-(i:Incident {crime_type:'fraud'})<-[r:SUSPECTED_OF INVESTIGATES]-(p:Person) RETURN DISTINCT p.name, type(r) <i>X wrong_result</i>
Q-056 Show people relevant to the Cannington homicide.	MATCH (i:Incident {crime_type:'homicide'})-[:OCCURRED_AT]->(l:Location {suburb:'Cannington'}) WITH i MATCH (p:Person)-[:SUSPECTED_OF WITNESSED VICTIM_OF INVESTIGATES]->(i) RETURN DISTINCT p.name <i>✓ matches default</i>	MATCH (i:Incident {crime_type:'homicide'})-[:OCCURRED_AT]->(l:Location {suburb:'Cannington'}) MATCH (p:Person)-[r]->(i) RETURN DISTINCT p.name <i>✓ matches default</i>
Q-057 Who is linked to the robbery in Midland?	MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'robbery'})-[:OCCURRED_AT]->(l:Location {suburb:'Midland'}) RETURN DISTINCT p.name <i>X valid_non_default</i>	MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'robbery'})-[:OCCURRED_AT]->(l:Location {suburb:'Midland'}) RETURN DISTINCT p.name <i>X valid_non_default</i>
Q-058 Who played a part in the homicide?	MATCH (p:Person)-[:SUSPECTED_OF WITNESSED VICTIM_OF INVESTIGATES]->(i:Incident {crime_type:'homicide'}) RETURN DISTINCT p.name <i>✓ matches default</i>	MATCH (p:Person)-[r]->(i:Incident {crime_type:'homicide'}) RETURN DISTINCT p.name <i>✓ matches default</i>
Q-059 List the people connected to the incidents in the Financial Crimes case.	MATCH (c:Case {case_name:'Financial Crimes'})-[:CONTAINS]->(i:Incident)<-[:SUSPECTED_OF]-(p:Person) RETURN DISTINCT p.name <i>X wrong_result</i>	MATCH (c:Case {case_name:'Financial Crimes'})-[:CONTAINS]->(i:Incident)<-[:SUSPECTED_OF]-(p:Person) RETURN DISTINCT p.name <i>X wrong_result</i>
Q-060 Who is involved in the burglary in Fremantle — suspects, witnesses, or the officer?	MATCH (i:Incident {crime_type:'burglary'})-[:OCCURRED_AT]->(l:Location {suburb:'Fremantle'}) WITH i OPTIONAL MATCH (suspect:Person)-[:SUSPECTED_OF]->(i) OPTIONAL MATCH (witness:Person)-[:WITNESSED]->(i) OPTIONAL MATCH (officer:Person)-[:INVESTIGATES]->(i) RETURN DISTINCT suspect.name AS suspect, witness.name AS witness, officer.name AS officer	MATCH (p:Person)-[:SUSPECTED_OF WITNESSED INVESTIGATES]->(i:Incident {crime_type:'burglary'})-[:OCCURRED_AT]->(l:Location {suburb:'Fremantle'}) RETURN DISTINCT p.name <i>X wrong_result</i>

Question	C2 — schema grounded	C3 — disambiguation enhanced
	<i>X wrong_result</i>	

Entity ambiguity

$n=20$ — 1 fixed by C3, 0 regressed, 6 both_correct, 13 both_wrong

Question	C2 — schema_grounded	C3 — disambiguation_enhanced
Q-061 What is Tran's current address?	<pre>MATCH (p:Person)-[:LIVES_AT {active:true}]->(l:Location) WHERE p.name CONTAINS 'Tran' RETURN l.address</pre> <i>✓ matches default</i>	<pre>MATCH (p:Person)-[:LIVES_AT {active:true}]->(l:Location) WHERE p.name CONTAINS 'Tran' RETURN l.address</pre> <i>✓ matches default</i>
Q-062 Where does Kim currently live?	<pre>MATCH (p:Person {name:'Kim'})-[:LIVES_AT {active:true}]->(l:Location) RETURN l.address</pre> <i>✗ wrong_result</i>	<pre>MATCH (p:Person {name:'Kim'})-[:LIVES_AT {active:true}]->(l:Location) RETURN l.address</pre> <i>✗ wrong_result</i>
Q-063 What is Nguyen's phone number?	<pre>MATCH (p:Person)-[:USES_PHONE]->(ph:Phone) WHERE p.name CONTAINS 'Nguyen' RETURN DISTINCT ph.phone_number</pre> <i>✗ wrong_result</i>	<pre>MATCH (p:Person)-[:USES_PHONE {active: true}]->(ph:Phone) WHERE p.name CONTAINS 'Nguyen' RETURN DISTINCT ph.phone_number</pre> <i>✓ matches default</i>
Q-064 Which incidents is Chen connected to?	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident) WHERE p.name CONTAINS 'Chen' RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status</pre> <i>✗ wrong_result</i>	<pre>MATCH (p:Person {name:'Chen'})-[:SUSPECTED_OF]->(i:Incident) RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status</pre> <i>✗ wrong_result</i>
Q-065 Which incidents is Smith connected to?	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident) WHERE p.name CONTAINS 'Smith' RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status</pre> <i>✗ wrong_result</i>	<pre>MATCH (p:Person {name:'Smith'})-[:SUSPECTED_OF]->(i:Incident) RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status</pre> <i>✗ wrong_result</i>
Q-066 Which incidents involve Bennett?	<i>✗ invalid_query</i>	<pre>MATCH (p:Person {name:'Bennett'})-[:SUSPECTED_OF]->(i:Incident) RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status</pre> <i>✗ wrong_result</i>
Q-067 Where does James live?	<pre>MATCH (p:Person {name:'James'})-[:LIVES_AT]->(l:Location) RETURN l.address</pre> <i>✗ wrong_result</i>	<pre>MATCH (p:Person {name:'James'})-[:LIVES_AT]->(l:Location) RETURN l.address</pre> <i>✗ wrong_result</i>
Q-068 Where does Sarah live?	<pre>MATCH (p:Person {name:'Sarah'})-[:LIVES_AT]->(l:Location) RETURN l.address</pre> <i>✗ wrong_result</i>	<pre>MATCH (p:Person {name:'Sarah'})-[:LIVES_AT {active:true}]->(l:Location) RETURN l.address</pre> <i>✗ wrong_result</i>
Q-069 Which incidents is David connected to?	<pre>MATCH (p:Person {name:'David'})-[:SUSPECTED_OF WITNESSED VICTIM_OF INVESTIGATES]->(i:Incident) RETURN DISTINCT i.incident_id</pre> <i>✗ wrong_result</i>	<pre>MATCH (p:Person {name:'David'})-[:SUSPECTED_OF WITNESSED VICTIM_OF INVESTIGATES]->(i:Incident) RETURN DISTINCT i.incident_id</pre> <i>✗ wrong_result</i>
Q-070 Which incident occurred on Albany Highway?	<pre>MATCH (i:Incident)-[:OCCURRED_AT]->(l:Location {address:'Albany Highway'}) RETURN i.incident_id</pre> <i>✗ wrong_result</i>	<pre>MATCH (i:Incident)-[:OCCURRED_AT]->(l:Location {address:'Albany Highway'}) RETURN i.incident_id</pre> <i>✗ wrong_result</i>
Q-071 Who lives on Albany Highway?	<pre>MATCH (p:Person)-[:LIVES_AT]->(l:Location {address:'Albany Highway'}) RETURN DISTINCT p.name</pre> <i>✗ wrong_result</i>	<pre>MATCH (p:Person)-[:LIVES_AT {active:true}]->(l:Location {address:'Albany Highway'}) RETURN DISTINCT p.name</pre> <i>✗ wrong_result</i>
Q-072 What incidents is Tran involved in?	<i>✗ invalid_query</i>	<i>✗ invalid_query</i>
Q-073 What is Nguyen's current address?	<pre>MATCH (p:Person)-[:LIVES_AT {active:true}]->(l:Location) WHERE p.name CONTAINS 'Nguyen' RETURN l.address</pre>	<pre>MATCH (p:Person)-[:LIVES_AT {active:true}]->(l:Location) WHERE p.name CONTAINS 'Nguyen' RETURN l.address</pre>

Question	C2 — schema grounded	C3 — disambiguation enhanced
	✓ matches default	✓ matches default
Q-074 Which incidents is Kim connected to?	MATCH (p:Person {name:'Kim'})-[:ASSOCIATED_WITH]-(p2:Person)-[:SUSPECTED_OF]->(i:Incident) RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status ✗ wrong_result	MATCH (p:Person {name:'Kim'})-[:SUSPECTED_OF]->(i:Incident) RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status ✗ wrong_result
Q-075 Which incidents is Sarah connected to?	MATCH (p:Person {name:'Sarah'})-[:SUSPECTED_OF WITNESSED VICTIM_OF INVESTIGATES]->(i:Incident) RETURN DISTINCT i.incident_id ✗ wrong_result	MATCH (p:Person {name: 'Sarah Bennett'})-[r]->(i:Incident) RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status ✗ wrong_result
Q-076 Show the robbery incident.	MATCH (i:Incident {crime_type:'robbery'}) RETURN i.incident_id, i.crime_type, i.date, i.status ✗ wrong_result	MATCH (i:Incident {crime_type: 'robbery'}) RETURN i.incident_id, i.crime_type, i.date, i.status ✗ wrong_result
Q-077 Who is suspected of the robbery?	MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'robbery'}) RETURN DISTINCT p.name ✓ matches default	MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'robbery'}) RETURN DISTINCT p.name ✓ matches default
Q-078 Who witnessed the assault?	MATCH (p:Person)-[:WITNESSED]->(i:Incident {crime_type:'assault'}) RETURN DISTINCT p.name ✓ matches default	MATCH (p:Person)-[:WITNESSED]->(i:Incident {crime_type:'assault'}) RETURN DISTINCT p.name ✓ matches default
Q-079 Where did the fraud occur?	MATCH (i:Incident {crime_type:'fraud'})-[:OCCURRED_AT]->(l:Location) RETURN l.address ✓ matches default	MATCH (i:Incident {crime_type:'fraud'})-[:OCCURRED_AT]->(l:Location) RETURN l.address ✓ matches default
Q-080 Who investigated the drug offence?	MATCH (p:Person)-[:INVESTIGATES]->(i:Incident {crime_type:'drug_offence'}) RETURN p.name ✓ matches default	MATCH (p:Person)-[:INVESTIGATES]->(i:Incident {crime_type:'drug_offence'}) RETURN p.name ✓ matches default

Intent ambiguity

$n=20$ — 0 fixed by C3, 0 regressed, 7 both_correct, 13 both_wrong

Question	C2 — schema_grounded	C3 — disambiguation_enhanced
Q-081 Tell me about the suspects in the Lake Street drug offence.	<pre>MATCH (l:Location {address:'Lake Street'})<-[:OCCURRED_AT]-(i:Incident {crime_type:'drug_offence'})<-[:SUSPECTED_OF]-(p:Person) RETURN DISTINCT p.name, p.alias, p.date_of_birth, p.gender</pre> <i>X wrong_result</i>	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'drug_offence'})-[:OCCURRED_AT]->(l:Location {address:'Lake Street'}) RETURN DISTINCT p.person_id, p.name, p.alias, p.date_of_birth, p.gender</pre> <i>X wrong_result</i>
Q-082 What about the incidents in Northbridge?	<pre>MATCH (i:Incident)-[:OCCURRED_AT]->(l:Location {suburb:'Northbridge'}) RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status</pre> <i>X wrong_result</i>	<pre>MATCH (i:Incident)-[:OCCURRED_AT]->(l:Location {suburb:'Northbridge'}) RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status</pre> <i>X wrong_result</i>
Q-083 Show the current residents of 55 Beaufort Street.	<pre>MATCH (p:Person)-[:LIVES_AT {active:true}]->(l:Location {address:'55 Beaufort Street'}) RETURN DISTINCT p.name</pre> <i>✓ matches default</i>	<pre>MATCH (p:Person)-[:LIVES_AT {active:true}]->(l:Location {address:'55 Beaufort Street'}) RETURN DISTINCT p.name</pre> <i>✓ matches default</i>
Q-084 Show incidents in Perth.	<pre>MATCH (i:Incident)-[:OCCURRED_AT]->(l:Location {suburb:'Perth'}) RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status</pre> <i>X wrong_result</i>	<pre>MATCH (i:Incident)-[:OCCURRED_AT]->(l:Location) WHERE l.location_id IN ['LOC-005', 'LOC-008', 'LOC-014'] RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status</pre> <i>X wrong_result</i>
Q-085 Who is the main suspect in the homicide?	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'homicide'}) RETURN p.name</pre> <i>✓ matches default</i>	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'homicide'}) RETURN p.name</pre> <i>✓ matches default</i>
Q-086 How connected is Anton Maric?	<pre>MATCH (p:Person {name:'Anton Maric'})-[:ASSOCIATED_WITH]-(p2:Person) RETURN count(DISTINCT p2)</pre> <i>X wrong_result</i>	<pre>MATCH (p:Person {name: 'Anton Maric'})-[*1..3]-(other:Person) RETURN DISTINCT other.name</pre> <i>X wrong_result</i>
Q-087 List the calls made by James Whitfield's phone.	<pre>MATCH (per:Person {name:'James Whitfield'})-[:USES_PHONE]->(ph:Phone)-[:CALLED]->(ph2:Phone) RETURN ph2.phone_number</pre> <i>✓ matches default</i>	<pre>MATCH (p:Person {name:'James Whitfield'})-[:USES_PHONE]->(ph:Phone)-[:CALLED]->(ph2:Phone) RETURN DISTINCT ph2.phone_number</pre> <i>✓ matches default</i>
Q-088 How many people did Ricky Santos's phone contact?	<pre>MATCH (per:Person {name:'Ricky Santos'})-[:USES_PHONE]->(ph:Phone)-[:CALLED]->(ph2:Phone)<-[:USES_PHONE]-(per2:Person) RETURN COUNT(DISTINCT per2)</pre> <i>✓ matches default</i>	<pre>MATCH (per:Person {name:'Ricky Santos'})-[:USES_PHONE]->(ph:Phone)-[:CALLED]->(ph2:Phone)<-[:USES_PHONE]-(per2:Person) RETURN COUNT(DISTINCT per2)</pre> <i>✓ matches default</i>
Q-089 Tell me about the vehicles owned by the homicide suspects.	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'homicide'}), (p)-[:OWNS]->(v:Vehicle) RETURN DISTINCT v.plate, v.make, v.model, v.colour</pre> <i>X wrong_result</i>	<pre>MATCH (p:Person)-[:SUSPECTED_OF]->(i:Incident {crime_type:'homicide'}), (p)-[:OWNS]->(v:Vehicle) RETURN DISTINCT v.plate, v.make, v.model, v.colour</pre> <i>X wrong_result</i>
Q-090 What is the biggest case?	<pre>MATCH (c:Case)-[:CONTAINS]->(i:Incident) WITH c, count(i) AS incident_count ORDER BY incident_count DESC LIMIT 1 RETURN c.case_name</pre> <i>✓ matches default</i>	<pre>MATCH (c:Case)-[:CONTAINS]->(i:Incident) WITH c, count(i) AS incident_count ORDER BY incident_count DESC LIMIT 1 RETURN c.case_name</pre> <i>✓ matches default</i>
Q-091 How many incidents has Officer Sharma investigated?	<pre>MATCH (p:Person {name:'Officer Sharma'})-[:INVESTIGATES]->(i:Incident) RETURN count(i)</pre> <i>X wrong_result</i>	<pre>MATCH (p:Person {name: 'Officer Sharma'})-[:INVESTIGATES]->(i:Incident) RETURN count(i)</pre> <i>X wrong_result</i>

Question	C2 — schema grounded	C3 — disambiguation enhanced
Q-092 Show recent incidents in Northbridge.	MATCH (i:Incident)-[:OCCURRED_AT]->(l:Location {suburb:'Northbridge'}) RETURN i.incident_id, i.crime_type, i.date, i.status ORDER BY i.date DESC <i>X wrong_result</i>	MATCH (i:Incident)-[:OCCURRED_AT]->(l:Location {suburb:'Northbridge'}) RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status <i>X wrong_result</i>
Q-093 How many people has Anton Maric's phone been in contact with?	MATCH (person:Person {name:'Anton Maric'})-[:USES_PHONE]->(ph:Phone)-[:CALLED]-(otherPh:Phone)<-[:USES_PHONE]-(otherPerson:Person) RETURN COUNT(DISTINCT otherPerson) <i>X wrong_result</i>	MATCH (p:Person {name: 'Anton Maric'})-[:USES_PHONE]->(ph1:Phone)-[:CALLED]->(ph2:Phone)<-[:USES_PHONE]-(other:Person) RETURN count(DISTINCT other) <i>X wrong_result</i>
Q-094 Who is the primary investigating officer on the Hay Street fraud?	MATCH (c:Case {case_name:'Hay Street fraud'})-[:CONTAINS]->(i:Incident)<-[:INVESTIGATES]-(p:Person) RETURN p.name <i>X wrong_result</i>	MATCH (c:Case {case_name:'Hay Street fraud'})-[:CONTAINS]->(i:Incident)<-[:INVESTIGATES]-(p:Person) RETURN p.name <i>X wrong_result</i>
Q-095 Give me the witnesses to the homicide.	MATCH (p:Person)-[:WITNESSED]->(i:Incident {crime_type:'homicide'}) RETURN DISTINCT p.name <i>✓ matches default</i>	MATCH (p:Person)-[:WITNESSED]->(i:Incident {crime_type:'homicide'}) RETURN DISTINCT p.name <i>✓ matches default</i>
Q-096 List the incidents in Operation Ironside.	MATCH (c:Case {case_name:'Operation Ironside'})-[:CONTAINS]->(i:Incident) RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status <i>X wrong_result</i>	MATCH (c:Case {case_name: 'Operation Ironside'})-[:CONTAINS]->(i:Incident) RETURN DISTINCT i.incident_id, i.crime_type, i.date, i.status <i>X wrong_result</i>
Q-097 Show Daniel Kim's current phone.	MATCH (p:Person {name:'Daniel Kim'})-[:USES_PHONE {active:true}]->(ph:Phone) RETURN ph.phone_number <i>✓ matches default</i>	MATCH (p:Person {name:'Daniel Kim'})-[:USES_PHONE {active:true}]->(ph:Phone) RETURN ph.phone_number <i>✓ matches default</i>
Q-098 Which incidents happened in the city?	MATCH (i:Incident)-[:OCCURRED_AT]->(l:Location {suburb:'city'}) RETURN DISTINCT i.incident_id, i.crime_type, i.date <i>X wrong_result</i>	MATCH (i:Incident)-[:OCCURRED_AT]->(l:Location) RETURN DISTINCT i.incident_id <i>X valid_non_default</i>
Q-099 How busy is Officer Tran?	MATCH (p:Person {name:'Officer Tran'})-[:INVESTIGATES]->(i:Incident) RETURN count(i) <i>X wrong_result</i>	MATCH (p:Person {name: 'Officer Tran'})-[:INVESTIGATES]->(i:Incident) RETURN count(i) <i>X wrong_result</i>
Q-100 Who is the most-called person?	MATCH (caller:Phone)-[:CALLED]->(callee:Phone)<-[:USES_PHONE]-(person:Person) RETURN person.name, count(*) AS call_count ORDER BY call_count DESC LIMIT 1 <i>X wrong_result</i>	MATCH (caller:Phone)-[:CALLED]->(callee:Phone)<-[:USES_PHONE]-(p:Person) RETURN p.name, COUNT(*) AS call_count ORDER BY call_count DESC LIMIT 1 <i>X wrong_result</i>

Temporal ambiguity

$n=20$ — 13 fixed by C3, 0 regressed, 1 both_correct, 6 both_wrong

Question	C2 — schema_grounded	C3 — disambiguation_enhanced
Q-101 Where does James Whitfield live?	<pre>MATCH (p:Person {name:'James Whitfield'})-[:LIVES_AT]->(l:Location) RETURN l.address</pre> <i>X valid_non_default</i>	<pre>MATCH (p:Person {name:'James Whitfield'})-[:LIVES_AT {active:true}]->(l:Location) RETURN l.address</pre> <i>✓ matches default</i>
Q-102 Where does Lina Petrova live?	<pre>MATCH (p:Person {name:'Lina Petrova'})-[:LIVES_AT]->(l:Location) RETURN l.address</pre> <i>X valid_non_default</i>	<pre>MATCH (p:Person {name:'Lina Petrova'})-[:LIVES_AT {active:true}]->(l:Location) RETURN l.address</pre> <i>✓ matches default</i>
Q-103 What vehicle does Ricky Santos own?	<pre>MATCH (p:Person {name:'Ricky Santos'})-[:OWNS]->(v:Vehicle) RETURN v.make, v.model, v.plate, v.colour</pre> <i>X wrong_result</i>	<pre>MATCH (p:Person {name:'Ricky Santos'})-[:OWNS]->(v:Vehicle) RETURN v.make, v.model, v.plate, v.colour</pre> <i>X wrong_result</i>
Q-104 Who owns the Holden Commodore?	<pre>MATCH (p:Person)-[:OWNS]->(v:Vehicle {make:'Holden', model:'Commodore'}) RETURN p.name</pre> <i>X valid_non_default</i>	<pre>MATCH (p:Person)-[:OWNS]->(v:Vehicle {model:'Holden Commodore'}) RETURN DISTINCT p.name</pre> <i>X wrong_result</i>
Q-105 Who owns the Ford Ranger?	<pre>MATCH (p:Person)-[:OWNS]->(v:Vehicle {make:'Ford', model:'Ranger'}) RETURN DISTINCT p.name</pre> <i>X valid_non_default</i>	<pre>MATCH (p:Person)-[:OWNS {active:true}]->(v:Vehicle {make:'Ford', model:'Ranger'}) RETURN p.name</pre> <i>✓ matches default</i>
Q-106 What phone does Lina Petrova use?	<pre>MATCH (p:Person {name:'Lina Petrova'})-[:USES_PHONE]->(ph:Phone) RETURN ph.phone_number</pre> <i>X valid_non_default</i>	<pre>MATCH (p:Person {name:'Lina Petrova'})-[:USES_PHONE {active:true}]->(ph:Phone) RETURN ph.phone_number</pre> <i>✓ matches default</i>
Q-107 Who uses phone number 0412 345 678?	<pre>MATCH (p:Person)-[:USES_PHONE]->(ph:Phone {phone_number:'0412 345 678'}) RETURN p.name</pre> <i>X valid_non_default</i>	<pre>MATCH (p:Person)-[:USES_PHONE {active:true}]->(ph:Phone {phone_number: '0412 345 678'}) RETURN p.name</pre> <i>✓ matches default</i>
Q-108 Who uses the phone 0478 444 555?	<pre>MATCH (p:Person)-[:USES_PHONE]->(ph:Phone {phone_number:'0478 444 555'}) RETURN p.name</pre> <i>X valid_non_default</i>	<pre>MATCH (p:Person)-[:USES_PHONE {active:true}]->(ph:Phone {phone_number: '0478 444 555'}) RETURN p.name</pre> <i>✓ matches default</i>
Q-109 Who is Tyler Birch associated with?	<pre>MATCH (p:Person {name:'Tyler Birch'})-[:ASSOCIATED_WITH]->(p2:Person) RETURN DISTINCT p2.name</pre> <i>X wrong_result</i>	<pre>MATCH (p:Person {name:'Tyler Birch'})-[:ASSOCIATED_WITH]->(p2:Person) RETURN DISTINCT p2.name</pre> <i>X valid_non_default</i>
Q-110 Who is Amir Hassan associated with?	<pre>MATCH (p:Person {name:'Amir Hassan'})-[:ASSOCIATED_WITH]->(p2:Person) RETURN DISTINCT p2.name</pre> <i>X wrong_result</i>	<pre>MATCH (p:Person {name:'Amir Hassan'})-[:ASSOCIATED_WITH]->(other:Person) RETURN DISTINCT other.name</pre> <i>X valid_non_default</i>
Q-111 Where did Daniel Kim live?	<pre>MATCH (p:Person {name:'Daniel Kim'})-[:LIVES_AT]->(l:Location) RETURN l.address</pre> <i>X valid_non_default</i>	<pre>MATCH (p:Person {name:'Daniel Kim'})-[:LIVES_AT]->(l:Location) RETURN l.address</pre> <i>X valid_non_default</i>
Q-112 Where does Amir Hassan live?	<pre>MATCH (p:Person {name:'Amir Hassan'})-[:LIVES_AT]->(l:Location) RETURN l.address</pre> <i>X valid_non_default</i>	<pre>MATCH (p:Person {name:'Amir Hassan'})-[:LIVES_AT {active:true}]->(l:Location) RETURN l.address</pre> <i>✓ matches default</i>
Q-113 Where does Tyler Birch live?	<pre>MATCH (p:Person {name:'Tyler Birch'})-[:LIVES_AT]->(l:Location) RETURN l.address</pre> <i>X valid_non_default</i>	<pre>MATCH (p:Person {name:'Tyler Birch'})-[:LIVES_AT {active:true}]->(l:Location) RETURN l.address</pre> <i>✓ matches default</i>
Q-114 Where does Connor Doyle live?	<pre>MATCH (p:Person {name:'Connor Doyle'})-[:LIVES_AT]->(l:Location) RETURN l.address</pre>	<pre>MATCH (p:Person {name:'Connor Doyle'})-[:LIVES_AT {active:true}]->(l:Location) RETURN l.address</pre>

Question	C2 — schema grounded	C3 — disambiguation enhanced
	<i>✗ valid_non_default</i>	<i>✓ matches default</i>
Q-115 Where does Sarah Nguyen live?	MATCH (p:Person {name:'Sarah Nguyen'})-[:LIVES_AT]->(l:Location) RETURN l.address <i>✗ valid_non_default</i>	MATCH (p:Person {name:'Sarah Nguyen'})-[:LIVES_AT {active:true}]->(l:Location) RETURN l.address <i>✓ matches default</i>
Q-116 What vehicle does Nina Orlov own?	MATCH (p:Person {name:'Nina Orlov'})-[:OWNS]->(v:Vehicle) RETURN v.make, v.model, v.plate, v.colour <i>✗ wrong_result</i>	MATCH (p:Person {name:'Nina Orlov'})-[:OWNS]->(v:Vehicle) RETURN v.make, v.model, v.plate, v.colour <i>✗ wrong_result</i>
Q-117 Who owns the Nissan Navara?	MATCH (p:Person)-[:OWNS]->(v:Vehicle {make:'Nissan', model:'Navara'}) RETURN DISTINCT p.name <i>✗ valid_non_default</i>	MATCH (p:Person)-[:OWNS {active:true}]->(v:Vehicle {make:'Nissan', model:'Navara'}) RETURN p.name <i>✓ matches default</i>
Q-118 Who owns the BMW 3 Series?	MATCH (p:Person)-[:OWNS]->(v:Vehicle {make:'BMW', model:'3 Series'}) RETURN DISTINCT p.name <i>✗ valid_non_default</i>	MATCH (p:Person)-[:OWNS {active:true}]->(v:Vehicle {make:'BMW', model:'3 Series'}) RETURN p.name <i>✓ matches default</i>
Q-119 What phone does Michael Nguyen use?	MATCH (p:Person {name:'Michael Nguyen'})-[:USES_PHONE]->(ph:Phone) RETURN ph.phone_number <i>✗ valid_non_default</i>	MATCH (p:Person {name:'Michael Nguyen'})-[:USES_PHONE {active:true}]->(ph:Phone) RETURN ph.phone_number <i>✓ matches default</i>
Q-120 Who was Tyler Birch associated with in 2024?	MATCH (p:Person {name:'Tyler Birch'})-[r:ASSOCIATED_WITH]->(p2:Person) WHERE r.from_date <= date('2024-12-31') AND (r.to_date >= date('2024-01-01') OR r.to_date IS NULL) RETURN DISTINCT p2.name <i>✓ matches default</i>	MATCH (p:Person {name:'Tyler Birch'})-[r:ASSOCIATED_WITH]->(p2:Person) WHERE r.from_date <= date('2024-12-31') AND (r.to_date >= date('2024-01-01') OR r.to_date IS NULL) RETURN DISTINCT p2.name <i>✓ matches default</i>